AI-KF on Crazyflie

Estimating altitude without a height sensor

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Background

- The **AI-Kalman Filter** (Cellini et al. 2025) performs better than a standard Kalman filter for estimation.
- The **Crazyflie** is a small drone with the same sensors used in the AI-KF paper.
- This is a good opportunity to test the AI-KF on a real platform!

Project Goals

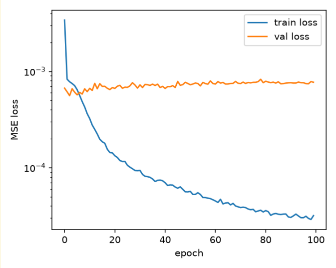
1. **Estimate on a collected trajectory** — run the AI-KF estimator on real, already-logged flight data and validate it against the withheld ToF reading.
2. **Real-time altitude estimation** — run the same estimator live on the Crazyflie [stretch goal]
3. **Feed the control strategy** — use the live estimate to actually inform how the drone is controlled, not just observe it. [stretch goal]

146 real flights, seven motifs

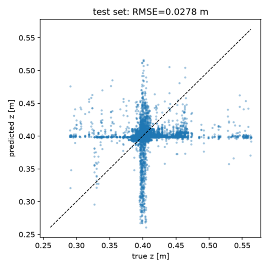
Motif	Reps	Why we flew it
accel_decel_pulse	96	The key motif — horizontal speed-up/slow-down is the only case the paper predicts makes altitude observable from flow + accel.
sum_of_sines	16	Continuous random-frequency acceleration, for frequency coverage beyond single pulses.
vertical_bob	8	Negative control — vertical-only motion, predicted near-zero observability.
offset_turn	8	Coordinated yaw + forward turn, to decouple heading from velocity direction.
zigzag	6	Task-representative coverage pattern flown at near-deployment speed.
mixed_free	6	Curated concatenation of motifs, closer to unstructured real flight.
hover	6	Baseline, low-observability control condition.

146 reps total, across ~15 flights · data_collection/trajectories.py

Trained directly on the real flights



TRAINING CURVE



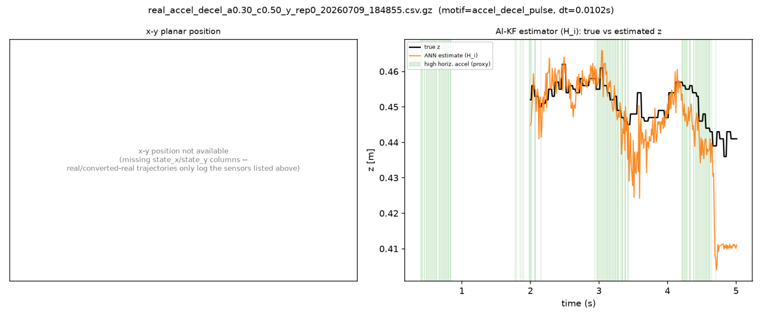
TEST PREDICTIONS

A 3×64 feed-forward net reads a sliding 2-second window (197 steps) of flow + accel and outputs altitude: **109** training reps, **30** held-out test reps, test **RMSE = 2.53 cm**.

Caveat

Encouraging, but not conclusive: nearly every flight targeted the **same nominal 0.4 m** altitude, so this mostly shows the estimator tracking small in-flight height jitter, not large deliberate height changes. More height variation in training data is next.

State estimate on a collected trajectory



Live estimation demo

NO SIGNAL · PENDING

Three directions from here

FUTURE WORK

- **Broaden the observability analysis** — we only recreated the sensor set from Ben's preprint. Worth running the same analysis on other sensors and states we haven't tried yet.
- **Scrutinize the drone model** — check whether our model is actually accurate, or can be improved, and tune the Kalman filter's $\mathbf{Q}$ and $\mathbf{R}$ matrices instead of leaving them as placeholders.
- **Improve training** — more height variation and more data.